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Technical Skills

Languages: Python, JavaScript (ES6+), SQL, C, HTML5, CSS3
Frontend: React.js, Tailwind CSS, Responsive Design, UI/UX Principles
Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, API Integration
Design Tools: Figma, Adobe Illustrator, Canva
Databases: MongoDB, MySQL, Firebase, Query Optimization, Database Normalization
DevOps & Cloud: Docker, Kubernetes, AWS (IAM, EC2, S3), CI/CD Pipelines (GitHub Actions)
ML / AI: Scikit-learn, Supervised Learning, Feature Engineering, PCA, Model Evaluation
Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Vercel, npm
Core CS: Data Structures & Algorithms, OOP, DBMS,

Professional Experience

Rashtriya Ispat Nigam Limited (RINL) — Vizag Steel Plant

Vizag, India

Database Systems Intern

Jan 2025 – Feb 2025

- Optimized 50+ complex SQL queries leveraging **JOIN operations, subqueries, indexing strategies, and GROUP BY aggregations**, reducing average query execution time by **40%** and significantly improving overall database throughput.
- Redesigned and normalized 10+ production relational database schemas to **Third Normal Form (3NF)**, cutting data redundancy by **30%** and enhancing long-term data integrity and maintainability.
- Managed and analyzed large-scale industrial datasets exceeding **100,000+ records** in a production RDBMS environment, generating actionable reports and insights that directly supported production planning and quality control decisions.

NIST University

Brahmapur, Odisha

Machine Learning Intern

Jun 2024 – Aug 2024

- Architected and deployed an **ML-based healthcare prediction system** using Python and Scikit-learn, achieving **92% prediction accuracy** across a validation dataset of 1,000+ patient records using supervised learning algorithms.
- Engineered a robust data pipeline encompassing preprocessing, feature engineering, and dimensionality reduction via **correlation analysis and PCA**, boosting model performance by **15%** and cutting training time by **25%**.
- Conducted rigorous model evaluation using **k-fold cross-validation, confusion matrix analysis, and ROC-AUC metrics**, validating model reliability and iteratively optimizing classification thresholds for production deployment.

Infosys Virtual Internship Program

Remote

Python Programming Intern

Jul 2023 – Sep 2023

- Completed 50+ structured coding exercises and mini-projects covering **Python fundamentals, OOP design patterns, and core data structures** (lists, dictionaries, sets, stacks), building a strong foundation in clean, modular programming.
- Implemented production-quality **file I/O handling, custom exception hierarchies, and systematic debugging workflows**, improving code reliability and reducing runtime error frequency across collaborative projects.

Projects

CLOTHSY | *React.js, Node.js, Express.js, MongoDB, JWT, REST API* [Live](#) [GitHub](#)

- Engineered a **production-grade full-stack e-commerce platform** end-to-end, featuring JWT-secured authentication, a dynamic shopping cart, and a **100+ product catalog** backed by real-time inventory management.
- Designed and implemented a **RESTful API layer** with optimized MongoDB aggregation pipelines, achieving sub-2-second page load times and **98% uptime** under concurrent user load through query indexing and efficient schema design.

Thumblify — AI Thumbnail Generator | *React.js, Node.js, REST API, AI Integration* [Live](#) [GitHub](#)

- Built and deployed a **full-stack AI-powered thumbnail generation application** on Vercel, integrating third-party generative AI APIs through a custom Node.js middleware layer with structured RESTful endpoints.
- Designed a **responsive, mobile-first React frontend** with clean state management and real-time preview capabilities, delivering an intuitive UX for content creators to generate and download AI thumbnails on demand.

Modern Developer Portfolio | *React.js, Tailwind CSS, JavaScript, Vercel* [Live](#) [GitHub](#)

- Developed a **high-performance personal portfolio** achieving a **95+ Google Lighthouse score** across all categories, leveraging code splitting, lazy loading, and mobile-first responsive design with fluid CSS animations.

Education

NIST University Brahmapur, Odisha
Bachelor of Technology in Computer Science and Engineering | **CGPA: 8.7 / 10.0** *Aug 2022 – May 2026*

Certifications

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| • Python Programming — Infosys | 2023 |
| • Machine Learning — NIST University | 2024 |
| • Database Management Systems — Rashtriya Ispat Nigam Limited (RINL) | 2025 |